

Assignment module – 2

1. Overview of c programming

the c programming language is one of the most important and influential programming language.

It was developed in the early 1970's by dennis Ritchie at bell labs.

C programming is important because it forms the functions for many famous programming languages like python, java, c++ .

Many concept like loops functions variable, and pointer used in modern language come directly from c.

The first major use of c was in rewriting the unix os since unix was written in c it became easier to move the os from one computer system to another computer system.

Another reason c remains popular is its educational value learning c helps student how computer works internally including memory management and data handling.

In conclusion c remains one of the most powerful and widely used programming language in the world...

Lab ex

1. Embedded system

- C is used in device like washing machine, cars, robots, and smart tv because it is fast and efficient.

2. Operating system

- Os like linux and windows use c for system level programming.

3. Game development

- C is used in game and graphics programming for high performance.

2. Setting up environment

1. Install gcc compiler
2. Add gcc bin path to environment variable
3. Check installation using

Ide options

Dev c++ - install and start coding

Codblocks - install codeBlocks with minGW setup

Visual studio code - install vs code add c extension and concept gcc compiler

Conclusion :-

Compiler converts c code into executable programs, and ides make coding easier

3.basic structure of c programming

A c program consists of different parts such as header files main function comments data types and variable.

Header file :- used to include in library functions examples include<stdio.h>

Main functions :- execution of program start from main()

Comments:- used to explain code

Data types:- defines the types of data like int float char

Variables:- used to store values in memory

Printf():- function display output on screen

Return 0:-indicates successful completion of program \

4. Operator in c

-  Arithmetic
-  Relational
-  Logical
-  Assignment
-  Increment/Decrement

- Bitwise
- Conditional (Ternary)
- Special Operators

1. arithmetic

Operator	Meaning	Example
+	Addition	$a + b$
-	Subtraction	$a - b$
*	Multiplication	$a * b$
/	Division	a / b
%	Modulus (remainder)	$a \% b$

2. Relational

Operator	Meaning
==	Equal to
!=	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal

3. Logical

Operator	Meaning
&&	Logical AND
,	
!	Logical NOT

4. Assignment

Operator	Example
=	$a = 5$
+=	$a += 2$

Operator Example

```
--      a  -=  2
*=      a  *=  2
/=      a  /=  2
```

5. Increment/Decrement

Operator Meaning

```
++      Increment
--      Decrement
```

6. Bitwise

Operator Meaning

```
&      Bitwise AND
^      XOR
~      NOT
<<     Left shift
>>     Right shift
```

7. Conditional (Ternary)

condition ? expression1 : expression2;

8. Special Operators

```
 sizeof
 & (address operator)
 * (pointer operator)
```